

PRACTICAL – 12

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PART 1: PRODUCER-CONSUMER PROBLEM

Aim

To implement the Producer-Consumer problem using counting semaphores and POSIX threads, and evaluate synchronization correctness and system throughput under different buffer sizes.

Theory

The Producer-Consumer problem involves a producer adding items to a shared buffer and a consumer removing them. Counting semaphores ensure that the producer does not add to a full buffer and the consumer does not remove from an empty buffer.

C Code

```
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>

#define BUFFER_SIZE 5
#define ITEMS 20

int buffer[BUFFER_SIZE];
int in = 0, out = 0;

sem_t empty, full;
pthread_mutex_t mutex;

void *producer(void *arg)
{
    for (int i = 1; i <= ITEMS; i++)
    {
        sem_wait(&empty);

        pthread_mutex_lock(&mutex);

        buffer[in] = i;
        printf("Produced: %d\n", i);
        in = (in + 1) % BUFFER_SIZE;
        pthread_mutex_unlock(&mutex);

        sem_post(&full);
    }
}
```

```

    }

    return NULL;
}

void *consumer(void *arg)
{
    for (int i = 1; i <= ITEMS; i++)
    {
        sem_wait(&full);

        pthread_mutex_lock(&mutex);

        int item = buffer[out];
        printf("Consumed: %d\n", item);
        out = (out + 1) % BUFFER_SIZE;

        pthread_mutex_unlock(&mutex);

        sem_post(&empty);
    }

    return NULL;
}

int main()
{
    pthread_t producer_thread, consumer_thread;

    sem_init(&empty, 0, BUFFER_SIZE);
    sem_init(&full, 0, 0);
    pthread_mutex_init(&mutex, NULL);

    pthread_create(&producer_thread, NULL, producer, NULL);
    pthread_create(&consumer_thread, NULL, consumer, NULL);

    pthread_join(producer_thread, NULL);
    pthread_join(consumer_thread, NULL);

    sem_destroy(&empty);
    sem_destroy(&full);
    pthread_mutex_destroy(&mutex);
    return 0;
}

```

OUTPUT –

```
teja@Hasini:~$ ./pc_demo
Produced: 1
Produced: 2
Produced: 3
Produced: 4
Produced: 5
Consumed: 1
Consumed: 2
Consumed: 3
Consumed: 4
Consumed: 5
Produced: 6
Produced: 7
Produced: 8
Produced: 9
Produced: 10
Consumed: 6
Consumed: 7
Consumed: 8
Consumed: 9
Consumed: 10
Produced: 11
Produced: 12
Produced: 13
Produced: 14
Produced: 15
Consumed: 11
Consumed: 12
Consumed: 13
Consumed: 14
Consumed: 15
Produced: 16
Produced: 17
Produced: 18
Produced: 19
Produced: 20
Consumed: 16
```

Compile

```
gcc producer_consumer.c -o pc -pthread
```

Run

```
./pc
```

Flow Chart

START

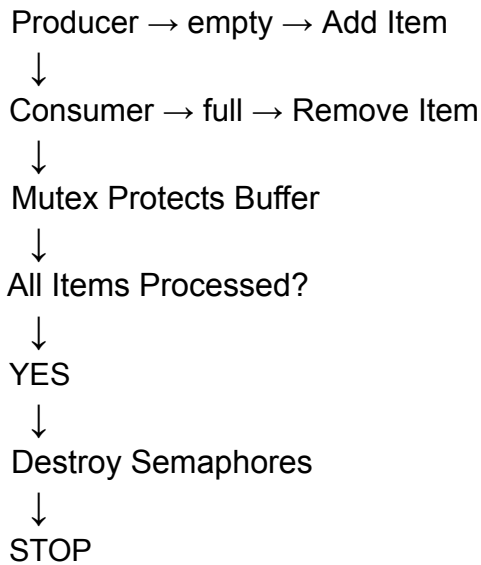


Initialize Buffer and Semaphores



Create Producer and Consumer





Buffer Size Evaluation

Buffer Size	Synchronization	Throughput
2	Correct	Lower
5	Correct	Good
10	Correct	Higher
20	Correct	Depends on system

Larger buffers can reduce waiting between producer and consumer, but the actual throughput depends on CPU, scheduling, and workload.

OUTPUT:

Result

The Producer-Consumer problem was successfully implemented using POSIX threads, counting semaphores, and a mutex. Synchronization was maintained correctly and buffer size affected throughput.

PART 2: DEADLOCK AND PREVENTION USING RESOURCE ORDERING

Aim

To design a deadlock scenario involving multiple threads competing for shared resources, demonstrate deadlock occurrence, identify the four necessary conditions for deadlock, and prevent deadlock using resource ordering.

Theory

A deadlock occurs when threads wait indefinitely for resources held by each

other.

The four necessary conditions are:

1. Mutual Exclusion – A resource can be used by only one thread at a time.
2. Hold and Wait – A thread holds one resource while waiting for another.
3. No Preemption – A resource cannot be forcibly taken from a thread.
4. Circular Wait – Threads form a circular chain of waiting.

Deadlock Code

```
#include <stdio.h>
#include <pthread.h>
#include <unistd.h>

pthread_mutex_t resource1 = PTHREAD_MUTEX_INITIALIZER;
pthread_mutex_t resource2 = PTHREAD_MUTEX_INITIALIZER;

void *thread1(void *arg)
{
    pthread_mutex_lock(&resource1);
    printf("Thread 1 locked Resource 1\n");

    sleep(1);

    pthread_mutex_lock(&resource2);
    printf("Thread 1 locked Resource 2\n");

    pthread_mutex_unlock(&resource2);
    pthread_mutex_unlock(&resource1);

    return NULL;
}

void *thread2(void *arg)
{
    pthread_mutex_lock(&resource2);
    printf("Thread 2 locked Resource 2\n");

    sleep(1);

    pthread_mutex_lock(&resource1);
    printf("Thread 2 locked Resource 1\n");

    pthread_mutex_unlock(&resource1);
    pthread_mutex_unlock(&resource2);

    return NULL;
}
```

```

int main()
{
    pthread_t t1, t2;

    pthread_create(&t1, NULL, thread1, NULL);
    pthread_create(&t2, NULL, thread2, NULL);

    pthread_join(t1, NULL);
    pthread_join(t2, NULL);

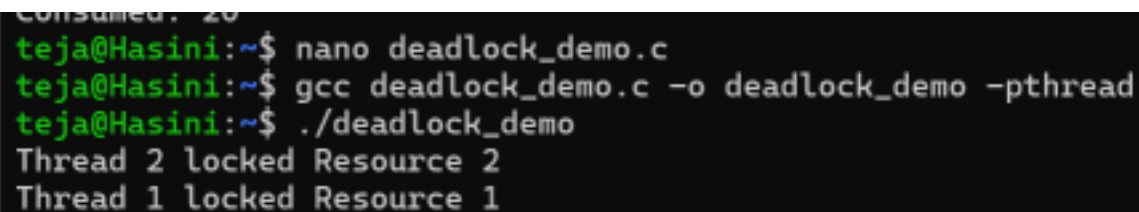
    return 0;
}

```

Compile

gcc deadlock.c -o deadlock -pthread

OUTPUT:



```

consumed: 20
teja@Hasini:~$ nano deadlock_demo.c
teja@Hasini:~$ gcc deadlock_demo.c -o deadlock_demo -pthread
teja@Hasini:~$ ./deadlock_demo
Thread 2 locked Resource 2
Thread 1 locked Resource 1

```

Run

./deadlock

The program may remain stuck because each thread waits for a resource held by the other.

PART 3: DEADLOCK PREVENTION USING RESOURCE ORDERING

Aim

To prevent deadlock by forcing all threads to acquire shared resources in the same order.

C Code

```

#include <stdio.h>
#include <pthread.h>
#include <unistd.h>

pthread_mutex_t resource1 = PTHREAD_MUTEX_INITIALIZER;
pthread_mutex_t resource2 = PTHREAD_MUTEX_INITIALIZER;

void *thread1(void *arg)
{
    pthread_mutex_lock(&resource1);
    printf("Thread 1 locked Resource 1\n");
}

```

```

    sleep(1);

    pthread_mutex_lock(&resource2);
    printf("Thread 1 locked Resource 2\n");

    pthread_mutex_unlock(&resource2);
    pthread_mutex_unlock(&resource1);

    return NULL;
}

void *thread2(void *arg)
{
    /* Same order: Resource 1 first, Resource 2 second */
    pthread_mutex_lock(&resource1);
    printf("Thread 2 locked Resource 1\n");

    sleep(1);

    pthread_mutex_lock(&resource2);
    printf("Thread 2 locked Resource 2\n");

    pthread_mutex_unlock(&resource2);
    pthread_mutex_unlock(&resource1);

    return NULL;
}

int main()
{
    pthread_t t1, t2;

    pthread_create(&t1, NULL, thread1, NULL);
    pthread_create(&t2, NULL, thread2, NULL);
    pthread_join(t1, NULL);
    pthread_join(t2, NULL);

    printf("No deadlock occurred.\n");

    return 0;
}

```

Compile

gcc prevention.c -o prevention -pthread

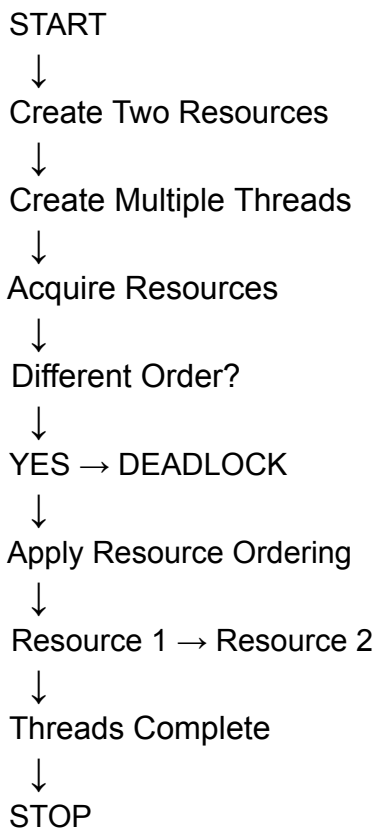
Run

./prevention

Expected Output

Thread 1 locked Resource 1
Thread 1 locked Resource 2
Thread 2 locked Resource 1
Thread 2 locked Resource 2
No deadlock occurred.

Flow Chart



Observation

In the deadlock program, each thread locks one resource and waits for the other, creating a circular wait. By forcing all threads to acquire resources in the same order, circular wait is eliminated.

OUTPUT:

```
teja@Hasini:~$ nano no_deadlock.c
teja@Hasini:~$ gcc no_deadlock.c -o no_deadlock -pthread
teja@Hasini:~$ ./no_deadlock
Thread 1 locked Resource 1
Thread 1 locked Resource 2
Thread 2 locked Resource 1
Thread 2 locked Resource 2
No deadlock occurred.
teja@Hasini:~$ |
```

Result

Deadlock was successfully demonstrated and prevented using resource ordering. The prevention strategy eliminates the circular-wait condition and allows all threads to complete successfully.